Surface-Code Syndrome Utilization for Fault Identification

Can repeated stabilizer syndromes reveal *where* a device is faulty — not just *that* it is?

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1. Motivation

Surface codes already measure syndromes every round — thousands of stabilizer bits are produced purely to detect and decode errors, then discarded.

> *"This syndrome stream is a free probe of the device. Can we read its own fault structure out of data we already collect for QEC?"*

Long-term vision: a **closed loop** that recycles QEC data into device knowledge, then feeds it back into better decoding.

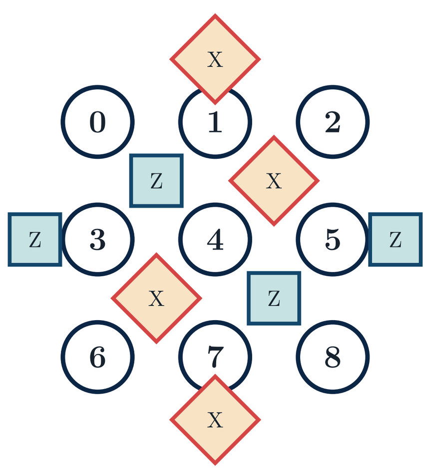
Repeated syndrome measurements
↓
Device information learning
↓
Fault / correlation structure inference
↓
Correlation- / device-aware decoding

feedback

This poster reports the **first link**: what fault information is — and is *not* — recoverable from syndromes alone.

2. Case Study & Problem

Setting. Distance-3 rotated surface code (17 qubits: 9 data + 8 ancilla); each stabilizer-extraction round issues **24 directed CNOTs**. One CNOT is the *dominant* fault source (higher fault probability than the rest).



Rotated d=3 layout (schematic): data qubits (circles), Z-/X-type ancillas (squares / diamonds).

The question

From a temporal syndrome / detection-event sequence $S = (s_1, \ldots, s_T)$ *alone*, identify the **dominant faulty CNOT** among the 24. **Nuisance factors** (unobserved): which rounds the fault fires, and its Pauli type $(XX, \ldots, ZZ)$.

3. Beyond Single-Shot

Exact enumeration (§5) bounds *single-shot* inference. Aggregating many noisy sequences / shots can push past that ceiling.

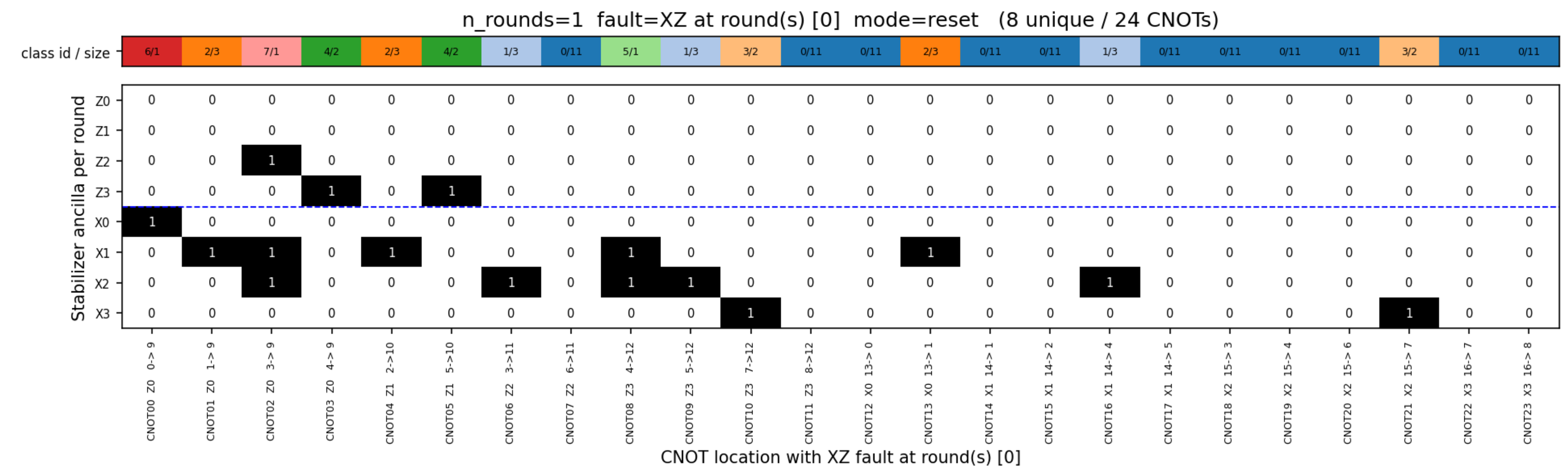
Preliminary, naive d=3 model

Rule-based single-round **ceiling**: $18/24 = 0.75$.
Sequence model (GRU), temporal: $\approx$95% .
Bag-of-shots (LogReg), spatial: $\approx$94% .

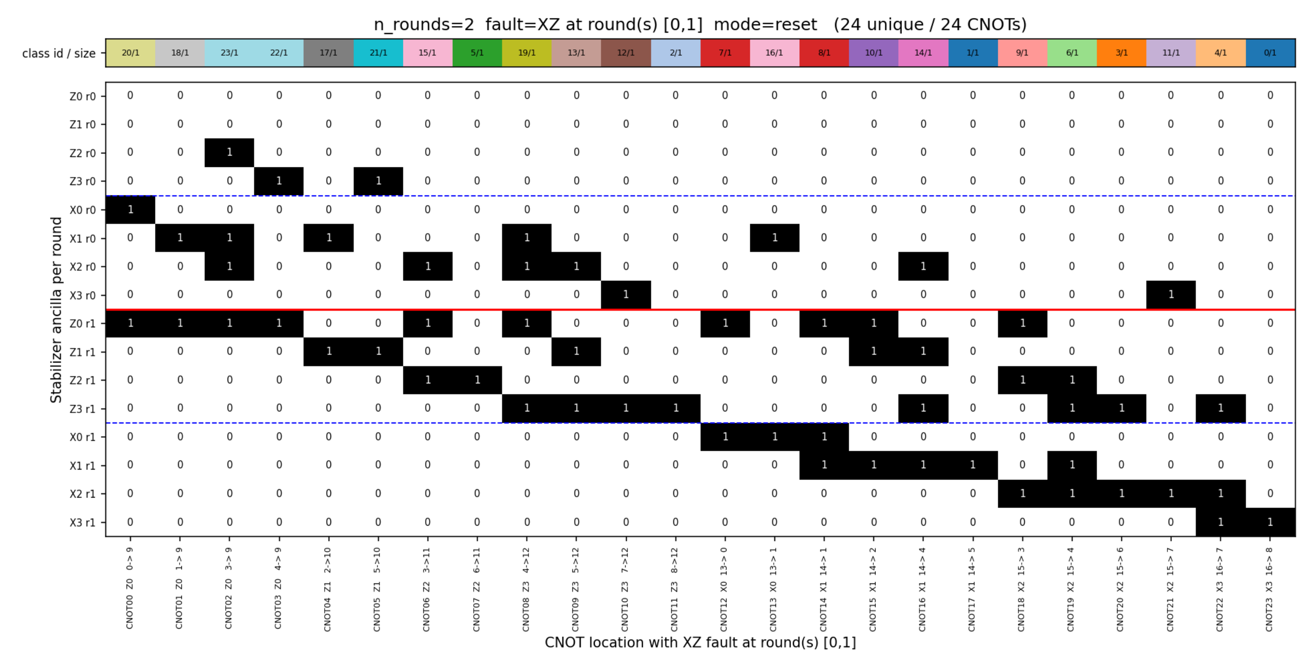
Statistical aggregation breaks the rule-based ceiling — reported as **preliminary**, not a final performance claim.

4. Temporal Information Threshold

Fix the Pauli type (XZ); enumerate the exact syndrome fingerprint of a single fault at each CNOT and count how many of the 24 become distinguishable.



One round. *Only 8 of 24 CNOTs are uniquely identified; 11 are syndrome-silent at round 0.*



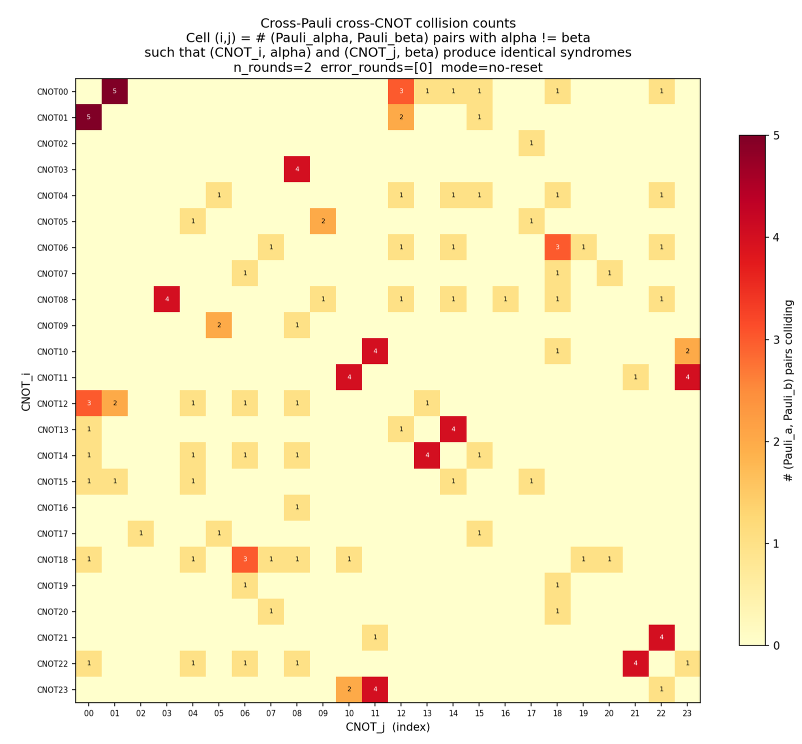
Two rounds. *24 of 24 CNOTs uniquely identified — a sharp threshold; further rounds add no new information.*

Takeaway

With the Pauli type known, the ***temporal* extent** of the record — not its size — unlocks identification: $1 \to 2$ rounds moves $8/24 \to 24/24$.

5. Cross-Pauli Ambiguity Wall

When the Pauli type is *unknown*, distinct (CNOT, Pauli) faults can produce *identical* syndromes. Enumerating all 216 cases:



Collision map: cell (i, j) counts Pauli pairs making faults at CNOT$_i$ and CNOT$_j$ indistinguishable (n_{rounds}=2).

Hard limit

The 216 cases collapse to 123 unique signatures with 72 cross-CNOT collisions. With single-round faults and unknown Pauli type, **single-shot 24-class identification is information-theoretically impossible** — invariant under round count and reset mode.

6. Status & Outlook

Done. Exact structural map of what syndromes can / cannot reveal about fault location at d=3.

Ongoing. Stim-based *realistic* circuit (parallel schedule, general d); *probabilistic* dominant-fault model.

Next. Close the loop — feed inferred fault / correlation structure into **device-aware decoding**.

Code, data & figures: `Correlated-Noise-Estimation` repository • Contact: pjhhan.research@gmail.com • Niels Bohr Quantum Summer School 2026